

Problem 8.12: Which three of the following four points lie on the same straight line:

$(32, 5)$

$(24, 18)$

$(22, 21)$

$(17, 29)$

8.6.5 Find a and b if the graph of $ax + 2y = b$ passes through $(2, 5)$ and is parallel to $3x - 5y = 7$.

8.42 Let P be $(-3, 7)$ and Q be $(5, -12)$.

(b) Find the point T on $\overline{PQ}$ such that $PT/TQ = 1/3$.

8.55 Suppose R is $(3, 5)$ and S is $(8, -3)$. Find each point on the line through R and S that is three times as far from R as it is from S . **Hints:** 161

8.57★ A line L has a slope of -2 and passes through the point $(r, -3)$. A second line, K , is perpendicular to L at (a, b) and passes through the point $(6, r)$. Find a in terms of r . **Hints:** 222

8.58★ How many points are common to the graphs of the two equations

$$(x - y + 2)(3x + y - 4) = 0 \quad \text{and} \quad (x + y - 2)(2x - 5y + 7) = 0?$$

8.59★

- (a) Given a real number r between 0 and 1 and two points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ in the plane, find the coordinates in terms of r of the point T on segment PQ such that $PT/PQ = r$.
- (b) What happens if you allow r to be greater than 1 in your formula in part (a)? Where on the line connecting P and Q is the resulting T ? Hints: 28
- (c) What happens if you allow r to be negative in your formula in part (a)? Where on the line connecting P and Q is the resulting T ? Hints: 171

9.2.1

- (b)★ Which number is larger, $\sqrt{3\sqrt{5}}$ or $\sqrt[3]{6\sqrt{2}}$? Hints: 41

9.2.3 List $\sqrt{2}$, $\sqrt[3]{3}$, and $\sqrt[5]{5}$ in order from least to greatest. (Source: Mandelbrot)

9.32 Find the maximum possible value for $x + y$, given that $3x + 2y \leq 7$ and $2x + 4y \leq 8$. (Source: Mandelbrot)

9.38 Describe all x such that $x^2 + 3 < 12$.

9.49★ Which is larger, 63^{45} or 33^{54} ? Hints: 150

Problem 10.25: Let a and b be the roots of the equation $x^2 - mx + 2 = 0$. Suppose that $a + (1/b)$ and $b + (1/a)$ are the roots of the equation $x^2 - px + q = 0$. What is q ? (Source: AMC 12)

10.4.5 The sum of the roots of a quadratic is $\frac{55}{72}$, and the product of the roots is $-\frac{25}{12}$. Find the roots.

Problem 10.27: For what values of k does the equation $\frac{x-1}{x-2} = \frac{x-k}{x-6}$ have no solution for x ? (Source: AMC 10)

10.40 Find all solutions to each equation below.

(d) $60 - 139r = 12r^2$

10.60★ Suppose the quadratic $x^2 + bx + c$ equals 0 when $x = r$ or $x = s$. If $r^2s + s^2r = 10$, and b and c are integers, find all possible ordered pairs (b, c) . **Hints:** 24

10.61★ Consider the quadratic $3y^2 - y - 12$.

10.62★ Suppose the quadratic has roots $y = r$ and $y = s$. Find a quadratic with roots $r + 2$ and $s + 2$.
Hints: 158

10.62★ At his usual rate a man rows 15 miles downstream in five hours less than it takes him to return. If he doubles his usual rate, the time downstream is only one hour less than the time upstream. What is the rate of the stream's current? (Source: AHSME) Hints: 216

10.63★ Find all triples (a, b, c) such that all three of the following equations are satisfied:

$$a(b + c - 5) = 7,$$

$$b(a + c - 5) = 7,$$

$$a^2 + b^2 = 50.$$